LAB 13 - PROMPTS

A/B-Test Stage: a Statistical Decision before Capital Moves

Open your AI coding assistant (for example Claude Code or GitHub Copilot) in the labs_fullcode folder. Work down the steps: copy each prompt into the assistant, review the code it writes, then run the command. The complete, working code is already in Lab13-ab-testing/solution/ if you want to compare. Never paste API keys into an AI chat.

Step 1: State the hypotheses (A5, K2)

Do this (no prompt needed)
Write H0: both variants have the same expected daily return and Sharpe; H1: they differ. Choose alpha = 0.05 and note how many variants the desk has tried this quarter (default 20).

Check: H0, H1, alpha and the number of variants tried are written down before the test runs.

Step 2: Build the statistics (A5, K2)

Prompt - copy into your AI coding assistant
Create labs_fullcode/pipeline/abtest.py. block_bootstrap_sharpe_diff(a, b, n_boot=2000, block=21, seed=7) resamples blocks of 21 consecutive days with numpy default_rng(seed) and returns Sharpe(A) - Sharpe(B) for each draw (annualised with sqrt(252)). ab_test(ra, rb, alpha=0.05, tried=20) returns the Welch t-test (scipy.stats.ttest_ind equal_var=False), the paired t-test (ttest_rel) and mean daily difference, the Sharpe difference and its (1 - alpha) bootstrap CI, the share of draws where A beats B, reject_h0 = paired p < alpha AND the CI excludes 0, and the Bonferroni threshold alpha / tried. Rules: run from the labs folder; reuse labs_fullcode/common (config, data, broker, indicators, metrics, backtest, plotting) and earlier pipeline stages instead of rewriting them; do not modify common/ or data/; read every setting from pipeline/config.yaml via pipeline.config.load_config(); never put API keys or secrets in code; paper trading only; keep a main(argv) function so `python -m pipeline <command>` works; add short comments a finance professional can follow.

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Check: ab_test returns a dict with welch_p, paired_p, ci and reject_h0.

Step 3: Build the A/B command (A5)

Prompt - copy into your AI coding assistant
In labs_fullcode/pipeline/abtest.py add run(a='trend_filter', b='dip_buy', alpha=0.05, tried=20, symbol='SPY') for `python -m pipeline abtest --a --b --alpha --tried --symbol` (a and b chosen from pipeline.signals.SIGNALS): backtest both with the config cost model, use the SAME window from the first valid 200-day SMA, print CAGR, volatility, Sharpe, max drawdown, exposure and trades, the three tests, a DECISION line ('REJECT H0 - ...' or 'DO NOT REJECT H0 - <winner> looks better on paper, but the evidence is not reliable; keep both on paper') and the Bonferroni caution; write _outputs/Lab13_abtest_decision.yaml and chart Lab13_ab_test.png (both equity curves and the bootstrap histogram with the CI). Rules: run from the labs folder; reuse labs_fullcode/common (config, data, broker, indicators, metrics, backtest, plotting) and earlier pipeline stages instead of rewriting them; do not modify common/ or data/; read every setting from pipeline/config.yaml via pipeline.config.load_config(); never put API keys or secrets in code; paper trading only; keep a main(argv) function so `python -m pipeline <command>` works; add short comments a finance professional can follow.

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Check: python -m pipeline abtest --help shows --a, --b, --alpha, --tried and --symbol.

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Step 4: Run the A/B test (A5)

Do this (no prompt needed)
Run: python -m pipeline abtest --a trend_filter --b dip_buy --alpha 0.05 (same as: python3 Lab13-ab-testing/abtest_stage.py).

Check: Window 2018-10-16 to 2025-12-31 (1,812 days). A trend_filter vs B dip_buy: CAGR 9.70% vs 4.97%, Sharpe 0.81 vs 0.49, max drawdown -19.36% vs -23.93%, trades 24 vs 102.

Step 5: Compare the independent and paired tests (K2)

Do this (no prompt needed)
Read tests 1 and 2. Why does the paired design suit two strategies measured on the same days?

Check: Welch t = 0.74 (p = 0.462); paired t = 0.85 (p = 0.398), mean difference 0.00018 a day.

Step 6: Read the bootstrap CI and the decision (A5)

Prompt - copy into your AI coding assistant
Rewrite this A/B test result for two portfolio managers who are not statisticians, in under 100 words: <paste the three test lines and the DECISION line>. Explain why 'A looks better' is not the same as 'A is better', and what evidence would change the decision.

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Check: Sharpe difference +0.31, 95% CI [-0.33, +1.05], A beats B in 82% of bootstrap samples; DECISION: DO NOT REJECT H0. Bonferroni with 20 variants: p < 0.0025.

Step 7: Test the pipeline's own strategy (A5, K2)

Do this (no prompt needed)
Run: python -m pipeline abtest --a ema_cross --b trend_filter.

Check: ema_cross Sharpe 0.95 vs trend_filter 0.81; Welch p 0.852, paired p 0.715, Sharpe difference +0.14, CI [-0.42, +0.67], A beats B in 66% - again DO NOT REJECT H0: the configured strategy is not proven better.

Run it

Step	Command / file
Run Lab 13	python3 Lab13-ab-testing/abtest_stage.py (same as: python -m pipeline abtest)
The complete code	pipeline/abtest.py (reference copy: Lab13-ab-testing/solution/)
Outputs	_outputs/ (charts, CSV files and reports written by the stage)

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